Topics 8.1 – 8.3: MCQ Practice

1. It is observed that the pH of pure water at a temperature of 37°C is equal to 6.81. Which of the following correctly identifies the pOH of pure water at 37°C and the value of K_w at 37°C?

	pOH of pure water at 37°C	K_w at 37°C
(A)	6.81	1.0×10^{-14}
(B)	6.81	2.4×10^{-14}
(C)	7.19	1.0×10^{-14}
(D)	7.19	2.4×10^{-14}

2. At 25°C, enough distilled water is added to a 100.0 mL sample of $HNO_3(aq)$ with a pH of 4.50 so that the final pH of the diluted solution is 5.20. The volume of distilled water added to the original solution is closest to

   (A) 16.0 mL

   (B) 116.0 mL

   (C) 400.0 mL

   (D) 500.0 mL

3. A student dissolves 2.0 grams of NaOH(*s*) in distilled water so that the total volume of the solution is 2.0×10^3 milliliters. The pH of the final solution at 25°C is closest to

   (A) 9.4

   (B) 12.4

   (C) 12.7

   (D) 14.0

Concentration (M)	pH of Acid 1	pH of Acid 2
5.0×10^{-3}	3.52	2.30
4.0×10^{-4}	4.07	3.40

4. The pH of solutions of two different monoprotic acids prepared at two different concentrations were measured and recorded in the table above. Which of the following statements correctly compares the relative strength of these acids and provides a valid justification?

   (A) Acid 1 is the stronger acid because it has a larger K_a value.

   (B) Acid 1 is the stronger acid because it has a smaller percent ionization value.

   (C) Acid 2 is the stronger acid because it has a smaller K_a value.

   (D) Acid 2 is the stronger acid because it has a larger percent ionization value.

$[HC_2H_3O_2]$ (M)	$[H_3O^+]$ (M)
1.0×10^{-2}	4.2×10^{-4}
1.0×10^{-3}	1.3×10^{-4}
1.0×10^{-4}	3.4×10^{-5}

5. The table above shows data for three different solutions of $HC_2H_3O_2(aq)$ at 25°C. Based on this data, which of the following is most likely to occur when a sample of $HC_2H_3O_2(aq)$ is diluted with water?

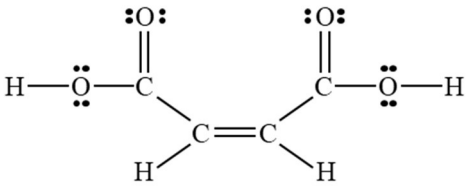
   (A) The value of K_a for $HC_2H_3O_2(aq)$ decreases.

   (B) The value of K_a for $HC_2H_3O_2(aq)$ increases.

   (C) The percent ionization of $HC_2H_3O_2(aq)$ decreases.

   (D) The percent ionization of $HC_2H_3O_2(aq)$ increases.

The structural formula of maleic acid ($H_2C_4H_2O_4$) is shown at right. Maleic acid is a diprotic acid, and the two acid ionization processes that occur are represented by the following equations.



$$H_2C_4H_2O_4(aq) + H_2O(l) \rightleftarrows HC_4H_2O_4^-(aq) + H_3O^+(aq) \qquad K_{a1} = 1.5 \times 10^{-2}$$

$$HC_4H_2O_4^-(aq) + H_2O(l) \rightleftarrows C_4H_2O_4^{2-}(aq) + H_3O^+(aq) \qquad K_{a2} = 8.5 \times 10^{-7}$$

6. Which of the following identifies the four species that are present in an aqueous solution of $H_2C_4H_2O_4$ from the highest to the lowest concentration at equilibrium?

(A) $H_2C_4H_2O_4 > HC_4H_2O_4^- > C_4H_2O_4^{2-} > H_3O^+$

(B) $H_2C_4H_2O_4 > H_3O^+ > HC_4H_2O_4^- > C_4H_2O_4^{2-}$

(C) $H_3O^+ > HC_4H_2O_4^- > C_4H_2O_4^{2-} > H_2C_4H_2O_4$

(D) $H_3O^+ > C_4H_2O_4^{2-} > HC_4H_2O_4^- > H_2C_4H_2O_4$

$HF(aq)$	$pK_a = 3.17$
$HCN(aq)$	$pK_a = 9.21$
$(CH_3)_3N(aq)$	$pK_b = 4.19$
$C_6H_5NH_2(aq)$	$pK_b = 9.42$

7. The pK_a or pK_b values of some weak acids and bases are listed in the table above. Which of the following solutions has the highest pH?

(A) 0.1 *M* $HF(aq)$

(B) 0.1 *M* $HCN(aq)$

(C) 0.1 *M* $(CH_3)_3N(aq)$

(D) 0.1 *M* $C_6H_5NH_2(aq)$

$$HOBr(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + OBr^-(aq) \qquad K_a = 2.2 \times 10^{-9}$$

8. The ionization of hypobromous acid, HOBr, is represented above. A certain solution of HOBr has a pH of 4.50. What is the approximate concentration of unionized HOBr(*aq*) in the solution?

   (A) 2.2×10^{-9} *M*

   (B) 3.2×10^{-5} *M*

   (C) 0.45 *M*

   (D) 4.5 *M*

9. Samples of $KNO_2(s)$ and $NH_4NO_3(s)$ are dissolved in separate beakers that each contain 100 mL of water. One of the salts produces a slightly basic solution. Which of the following equations best represents the formation of the slightly basic solution?

   (A) $K^+(aq) + 2\,H_2O(l) \rightleftarrows KOH(aq) + H_3O^+(aq)$

   (B) $NH_4^+(aq) + H_2O(l) \rightleftarrows NH_3(aq) + H_3O^+(aq)$

   (C) $NO_2^-(aq) + H_2O(l) \rightleftarrows HNO_2(aq) + OH^-(aq)$

   (D) $NO_3^-(aq) + H_2O(l) \rightleftarrows HNO_3(aq) + OH^-(aq)$

10. Caffeine ($C_8H_{10}N_4O_2$) is a weak base with a K_b value of 4.3×10^{-10}. The pH of a 0.01 *M* solution of caffeine is closest to

    (A) 5.7

    (B) 8.3

    (C) 9.4

    (D) 12.0

$$HC_3H_5O_2(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + C_3H_5O_2^-(aq) \qquad K_a = 1.3 \times 10^{-5}$$

11. Propanoic acid, $HC_3H_5O_2(aq)$, is a weak acid that ionizes in water according to the equation above. Which of the following provides the best estimate of the pH of 5×10^{-3} *M* $HC_3H_5O_2(aq)$ and identifies the species at the highest concentration (excluding H_2O) in the solution?

	pH	Species at Highest Concentration
(A)	2.3	$H_3O^+(aq)$
(B)	2.3	$HC_3H_5O_2(aq)$
(C)	3.6	$H_3O^+(aq)$
(D)	3.6	$HC_3H_5O_2(aq)$

$$OCl^-(aq) + H_2O(l) \rightleftarrows HOCl(aq) + OH^-(aq) \qquad K_b = ?$$

12. The hypochlorite ion, OCl^-, reacts with water according to the equation above. A solution of 0.025 *M* $NaOCl(aq)$ has a pH of 9.96. Which of the following is the best estimate of the value of the equilibrium constant, K_b, for the reaction shown above at 25°C?

(A) 4.8×10^{-19}

(B) 3.3×10^{-7}

(C) 9.1×10^{-5}

(D) 3.6×10^{-3}